

KENYA NATIONAL EXAMINATIONS COUNCIL

KCPE 2002

SCIENCE

READ THESE INSTRUCTIONS CAREFULLY

Time:1 hour 40 minutes

1. You have been given this question booklet and a separate answer sheet. The question booklet contains 50 question
2. Do any necessary rough work in this booklet.
3. When you have chosen your answer, mark it on the **ANSWER SHEET**, not in this question booklet.
4. Use only an ordinary pencil.
5. Make sure that you have written on the answer sheet:

YOUR INDEX NUMBER

YOUR NAME

NAME OF YOUR SHOOOL

6. By drawing a dark line inside the correct numbered boxes mark your full Index Number(i.e School code number and the three-figured Candidate's number)in the grid near the top of the answer sheet.
7. Do not make any marks outside the boxes.
8. Keep the sheet as clean as possible and do not fold it.
9. For each of the question 1-50 four answers are given. The answers are lettered A,B,C and D. In each case only ONE of the four answers is correct. Choose the correct answer.
10. On the answer sheet the correct answer is to be shown by drawing a dark line inside the box in which the letter have chosen is written.

Example

In the Question booklet.

4. Which one of the following components of blood is involved in the clotting after injury?

- A. Plasma
- B. White blood cells
- C. Red blood cells
- D. Platelets

The correct answer is D

1 [A] [B] [C] [D] 14. [A] [B] [C] [D] 24. [A] [B] [C] [D] 34. [A] [B] [C] [D]

11. Your dark line MUST be within the box.
12. For each question ONLY ONE box is to be marked in each set of four boxes.

This question paper consists of 7 printed pages and 1 black page.

The Kenya National Examination Council, 2002.

1. Which one of the following bodies produces light
 - A. Mars
 - B. Venus
 - C. Star
 - D. Moon
2. Which one of the following explains why flowering water moves objects
 - A. Has energy of movement
 - B. Is a liquid
 - C. Has potential energy
 - D. Carries less denser objects only
3. Which one of the following insects undergoes the stages of egg, Nymph and adult in its cycle?
 - A. Tsetse fly
 - B. Wasp
 - C. Butterfly
 - D. Cockroach
4. The diagram below represents parts of flower
The function of part labeled P is to
 - A. Produce nectar
 - B. Receives pollen grains
 - C. Lead pollen grain to ovary
 - D. Produce pollen grain
5. Engine oil mixes with kerosene but does not mix with water. Metylated spirit mixes with water but does not mix with kerosene. How many layers would be observed if the four liquids would be observed if the four liquids were put in the same container?
 - A. 1
 - B. 2
 - C. 3
 - D. 4
6. A chameleon shoots its tongue to
 - A. Taste food
 - B. Locate food
 - C. Catch food
 - D. Cool itself
7. The chart below represents a simple classification of plants
Which plant is represented by P and Q

P	Q
A. Ferns	moss
B. Banana	ferns
C. Moss	mushroom
D. Grass	banana

8. The diagram below represents parts of a germinating seed
The part labeled K
- A. Serves as first leaves
 - B. A becomes a shoot
 - C. Absorbs water for the young plant
 - D. Gives rise to root
9. On the cold day birds raises their feathers to
- A. Allow air to escape from their bodies
 - B. Allow even distribution of heat in the body
 - C. Avoid losing heat from their bodies
 - D. Generate heat
10. The diagram below represents an instruments that could be used to demonstrate that solid expand when heated
Which one of the following should be done to make the instrument more efficient?
- A. Using a thinner pointer
 - B. Reduce the length between support Q and pivot
 - C. Using thicker wire
 - D. Increase the length between P and pivot
11. Pupil placed little methylated spirit in a tin can and lit it. After a short while, they placed the tin can upside down over soft placed mud as shown in the diagram below.
The tin can sank into the soft mud. Which of the following explains why the tin can sank into the mud?
- A. The tin can is denser than soft mud
 - B. Air pressure on the tin can forced the tin can into the soft mud?
 - C. There was no air in the soft mud
 - D. Forced of gravity forced the tin can into the soft mud
12. Objects P, Q, R and S were balanced on a beam as shown in the diagram below
The correct order from the heaviest to the lightest objects is
- A. S Q R P
 - B. Q R P S
 - C. R S Q P
 - D. Q P S R
13. The diagram below represents a weather record that was made by pupils for 5 days
- Which one of the following statements about the weather is correct? Whenever it
- A. Was windy, there was rain
 - B. Was sunny, it was windy
 - C. Rained, it was windy
 - D. Was windy in the morning, it rained in the afternoon
14. Which one of diagram below represents the correct set up that is used to compare drainage of water in different types of soil

15. Which of the following drugs is a stimulant when used in small quantities but a depressant when in large quantities?
- A. Tobacco
 - B. Coffee
 - C. Khat
 - D. Alcohol
16. A stick was placed behind a bottle half filled with water as shown in the diagram below
The part of the stick below the level of water appeared to be larger in size than the part above the level of water because rays of light were
- A. Absorbed
 - B. Refracted
 - C. Dispersal
 - D. Reflected
17. Which one of the ways of producing light involves energy changes from chemical to electrical and lastly to heat and light? Using a
- A. Bicycle
 - B. Hydroelectric generator
 - C. Solar cells
 - D. Torch
18. The diagram below represents a mammalian heart name it correctly
19. Fruits which are dispersed by water
- A. Have pods with lines of weakness
 - B. Are small and hairy
 - C. Are hollow and fibrous
 - D. Are small with hooks
20. A certain animal lay eggs, has scale and the body temperature varies with the surrounding. The animals is likely to be a
- A. Snakes
 - B. Platypus
 - C. Toad
 - D. Duck
21. The type of modified root found in mangroves are called
- A. Breathing
 - B. Prop
 - C. Buttress
 - D. Clasping
22. Which one the following is the function of rectum?
- A. Completes digestion of food
 - B. Stores undigested matter
 - C. Absorbs water
 - D. Absorbs digested food

23. Which one of the following aspects of a given objects DOES NOT affect sinking or floating of the objects?
- Size
 - Material
 - Shape
 - Weight
24. A purple flower extract was added to substances KLM and N to find out whether they were acid or bases. The results are shown in the table below. Which of the two substances react to form salt water only?
- K and M
 - M and n
 - K and L
 - L and M
25. Which one the following aspects of an objects cannot be changed?
- Mass
 - Density
 - Weight
 - Volume
26. Which one of following should not be the same when comparing rates of melting in the substances
- Quantity of substance
 - Amount of heat
 - The substances
 - Duration of heat
27. Standard eight pupils suggested the following as a way of preventing monkeys from destroying crops
- Laying traps for the monkey
 - Putting up a fence around the farm
 - Seeking assistance of the game warden
 - Guarding the farm
- Which one of the following is not acceptable?
- (i)
 - (ii)
 - (iii)
 - (iv)
28. The diagram below represent lever being used used to open a tin
- Which one the following works in the same way as the lever shown above
- Carrying a load using a wheelbarrow
 - Cutting hair using using a scissors

- C. Scooping fish using a rod
 - D. Catching used
29. Which one the following statement about a single fixed pulley is not correct? The load
- A. Distance is equal to the effort distance
 - B. Moves in the opposite direction to that of the effort
 - C. Is twice the effort used
 - D. Is equal to the effort used
30. The planet with a ring around it is called
- A. Mercury
 - B. Saturn
 - C. Pluto
 - D. Jupiter
31. The soil that drains water fastest
- A. Has small air space
 - B. Cracks when dry
 - C. Has poor capillarity
 - D. Has smooth texture
32. The diagram below represents an improvised instrument that could be used to measure room temperature
- The working of the instrument is based on the fact that
- A. Liquids expands when heated and contract when cooled
 - B. Liquids rise in narrow tube
 - C. Gases exerts pressure
 - D. Gases expand when heated and contract when cooled
33. The component of air in the atmosphere whose amount varies from time is
- A. Oxygen
 - B. Water vapour
 - C. Carbon dioxide
 - D. Nitrogen
34. The stage involved in the formation of rain are
- I) Clouds are formed
 - II) Water evaporates
 - III) Heat from the sun warms water
 - IV) Rain falls
 - V) Water vapour condense

The correct order in which the stage takes place is

- A. (iii) (ii) (i) (v) (iv)
 - B. (ii) (iii) (v) (iv)(i)
 - C. (ii) (v) (iii) (iv)
 - D. (iii) (ii) (v) (i) (iv)
35. The diagram below represents an arrangement of bottle tops to show the working of gears

In which direction will F G and H turn when E is turned in a clockwise direction?

36. The diagram below represents a simple machines that is used to draw water from a river

The effort distance is

- A. XY
- B. WX
- C. YW
- D. YZ

37. The following activities involved in making an improvised hammer

- i) Making a hole
- ii) Fitting a bolt
- iii) Shaping a piece of wood
- iv) Making the handle smooth
- v) Fixing the nut

The correct order is

- A. (iii) (iv) (i) (ii) (v)
- B. (i) (ii) (v) (iii) (iv)
- C. (iii) (i) (ii) (v) (iv)
- D. (v) (ii) (iii) (iv) (i)

38. Which one of the following groups of drugs are harmful but legal in Kenya

- A. Tobacco, khat, beer
- B. Heroin, bhang, beer
- C. Bhang, heroin, mandrax
- D. Tobacco, mandax, khat

39. Vaccine are

- A. Stimulus
- B. Preventive drugs
- C. Curative drug
- D. Pain relievers

40. The planet that takes the longest time to move round the sun once is called

- A. Uranus
- B. Venus
- C. Mars
- D. Saturn

41. One of the following practices does not help conservation water. Which one is it

- A. Drip irrigation
- B. Using oil to kill mosquito larva
- C. Treating water from a factory
- D. Mulching

42. Birds which feed on grains have beaks that are

- A. Strong, short and curved

- B. Strong, short and cupped
 - C. Long, sharp and straight
 - D. Strong, short and straight
43. Which one of the following sets of characteristic are description of wind pollinated flower
- A. Fewer large pollen grains, petals sometimes absents
 - B. Fewer large pollen grains, large loosely attached anthers
 - C. Small in size, large loosely attached anthers stickly stigma
 - D. Small in size, large loosely attached anther petals sometimes absent
44. The diagram below represents a periscope and the position of the image at point U

The object is at point

- A. S
 - B. T
 - C. U
 - D. V
45. If the hole of the pinhole camera is widened the image
- A. Becomes blurred
 - B. Becomes larger
 - C. Is upright
 - D. Is not formed on the screen
46. Which one of the following pairs consists substances that have definite volume?
- A. Water vapour and ice
 - B. Sand and water
 - C. Stone and air
 - D. Ice and air
47. A suitable way of handling plastic materials to avoid pollution is
- A. Dumping in the river
 - B. Burning
 - C. Burying in the soil
 - D. Re-using
48. The diagram below represents a simple machine
- The machine is an example of
- A. Gear
 - B. Lever
 - C. Inclined plane
 - D. Wheel and axle
49. A group of people filled a transparent bottle with soil and water. They shook the bottle and left to stand for sometimes. The layer formed as shown in the diagram below
- The layer that is composed of water making
- A. 1
 - B. 2
 - C. 3

D. 4

50. Non-conductors of electricity can be used in making

- A. Insulators
- B. Handles of cooking
- C. Fuses
- D. Contact in switches